

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-E2E-001

Related Functional Requirement(s): FR-E2E-001

Objective: Implement a data ingestion job to read address change candidate records from Oracle staging source table.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Oracle JDBC driver, Spark Oracle connector

Source Data Details:

- Dataset Name: STG_E2E_AC_TXDBH_DATA
- Location: Oracle staging source table
- Format: Relational database table
- Description: Address change candidate records

Target Data Details:

- Output Name: Address Change Candidate Records
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Job Flow / Pipeline Stages:

- Read data from Oracle staging source table
- Pass data to next transformation stage

Data Transformations / Business Logic:

- Step: Read data from Oracle staging source table

- Description: Read address change candidate records from Oracle staging source table
- Transformation Logic: Use Oracle JDBC driver and Spark Oracle connector to read data from Oracle staging source table

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement retry mechanism for Oracle database connection failures

TR-E2E-002**Related Functional Requirement(s): FR-E2E-002**

Objective: Implement a data transformation job to create flat file output for DP Address Change YUYU.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark CSV writer

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: DP Address Change YUYU Flat File
- Location: Designated output directory
- Format: CSV
- Description: Flat file output containing required fields for DP Address Change YUYU

Job Flow / Pipeline Stages:

- Generate sequence number using SEQTRANS.NEXTVAL
- Derive required fields based on input data and business rules
- Create flat file output with required fields

Data Transformations / Business Logic:

- Step: Derive required fields
- Description: Derive required fields (e.g., ORIGIN_REQUEST_ID, POLICY_ID, RECEPTION_DATE, etc.) based on input data and business rules
- Transformation Logic: Use Spark DataFrame API to derive required fields

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-003**Related Functional Requirement(s): FR-E2E-003**

Objective: Implement a data update job to update staging table with process metadata and key identifiers.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Oracle JDBC driver, Spark Oracle connector

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Updated Staging Table
- Location: Oracle staging source table
- Format: Relational database table
- Description: Updated staging table with process metadata and key identifiers

Job Flow / Pipeline Stages:

- Update staging table with required fields based on input data and business rules

- Use DD_UPDATE update strategy to update table

Data Transformations / Business Logic:

- Step: Update staging table
- Description: Update staging table with required fields (e.g., STG_PROCESS_DATE, STG_PROCESS_USERID, etc.) based on input data and business rules
- Transformation Logic: Use Oracle JDBC driver and Spark Oracle connector to update staging table

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement retry mechanism for Oracle database connection failures

TR-E2E-004

Related Functional Requirement(s): FR-E2E-004

Objective: Implement a data transformation job to derive reception date in required format.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark DateTime API

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Derived Reception Date
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Derived reception date in required format

Job Flow / Pipeline Stages:

- Convert T_TX_REQUEST_REQ_ACC_DATETIME field to required format using specified business rule (BR-03)

Data Transformations / Business Logic:

- Step: Derive reception date
- Description: Derive reception date in required format based on input data and business rule (BR-03)
- Transformation Logic: Use Spark DateTime API to derive reception date

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-005

Related Functional Requirement(s): FR-E2E-005

Objective: Implement a data transformation job to apply address fallback logic when ZIP is missing.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark DataFrame API

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Applied Address Fallback Logic
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Applied address fallback logic when ZIP is missing

Job Flow / Pipeline Stages:

- Check if ZIP code is missing
- Apply fallback logic to derive required address fields based on specified business rule (BR-04)

Data Transformations / Business Logic:

- Step: Apply address fallback logic
- Description: Apply address fallback logic when ZIP is missing based on input data and business rule (BR-04)
- Transformation Logic: Use Spark DataFrame API to apply address fallback logic

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-006

Related Functional Requirement(s): FR-E2E-006

Objective: Implement a data transformation job to normalize postal code by removing hyphens.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark String API

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Normalized Postal Code
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Normalized postal code by removing hyphens

Job Flow / Pipeline Stages:

- Remove hyphens from postal code using specified business rule (BR-05)

Data Transformations / Business Logic:

- Step: Normalize postal code
- Description: Normalize postal code by removing hyphens based on input data and business rule (BR-05)
- Transformation Logic: Use Spark String API to normalize postal code

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-007**Related Functional Requirement(s): FR-E2E-007**

Objective: Implement a data transformation job to format telephone number according to specified rules.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark String API

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Formatted Telephone Number
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Formatted telephone number according to specified rules

Job Flow / Pipeline Stages:

- Format telephone number using specified business rule (BR-06)

Data Transformations / Business Logic:

- Step: Format telephone number
- Description: Format telephone number according to specified rules based on input data and business rule (BR-06)
- Transformation Logic: Use Spark String API to format telephone number

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-008**Related Functional Requirement(s): FR-E2E-008**

Objective: Implement a data transformation job to derive policy owner name in Kana and Kanji.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Spark DataFrame API, lookup tables

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Derived Policy Owner Name
- Location: Next transformation stage
- Format: Spark DataFrame
- Description: Derived policy owner name in Kana and Kanji

Job Flow / Pipeline Stages:

- Perform lookups on T_YUYU_CLNT and T_YUYUK_CLN tables to retrieve required data
- Derive policy owner name in Kana and Kanji using specified business rules (BR-07 and BR-08)

Data Transformations / Business Logic:

- Step: Derive policy owner name
- Description: Derive policy owner name in Kana and Kanji based on input data and business rules (BR-07 and BR-08)
- Transformation Logic: Use Spark DataFrame API and lookup tables to derive policy owner name

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement data quality checks to ensure output data accuracy

TR-E2E-009**Related Functional Requirement(s): FR-E2E-009**

Objective: Implement a data update job to stamp process metadata on staging update.

Target Cluster Configuration:

- Cluster Name: E2E Cluster
- Databricks Runtime Version: 7.3 LTS
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: Oracle JDBC driver, Spark Oracle connector

Source Data Details:

- Dataset Name: Address Change Candidate Records
- Location: Previous transformation stage
- Format: Spark DataFrame
- Description: Processed address change candidate records

Target Data Details:

- Output Name: Updated Staging Table with Process Metadata
- Location: Oracle staging source table
- Format: Relational database table
- Description: Updated staging table with process metadata

Job Flow / Pipeline Stages:

- Derive process metadata using specified business rules (BR-01 and BR-02)
- Update staging table with process metadata

Data Transformations / Business Logic:

- Step: Stamp process metadata
- Description: Stamp process metadata on staging update based on input data and business rules (BR-01 and BR-02)
- Transformation Logic: Use Oracle JDBC driver and Spark Oracle connector to update staging table with process metadata

Error Handling and Logging:

- Log errors and exceptions to a log file
- Implement retry mechanism for Oracle database connection failures